

**Tutorial 5: Second-Order Linear ODEs (Part II)**

[Solution]

**A) Cauchy-Euler Equations**Solve the given Euler equations on  $(0, \infty)$  :

$$(i) \quad x^2 y'' - 3xy' + 4y = 0, \quad (ii) \quad x^2 y'' - xy' - 35y = 0.$$

**Solution:** i)  $y(x) = c_1 x^2 + c_2 x^2 \ln x$    ii)  $y(x) = c_1 x^7 + c_2 x^{-5}$

**B) Method of Undetermined Coefficients**

1) Solve the initial value problem

$$y'' - y' - 2y = 10 \sin x, \quad y(0) = 0, \quad y'(0) = 1.$$

**Solution:**  $y(x) = e^{2x} - 2e^{-x} - 3 \sin x + \cos x.$

2) In each case, write an appropriate trial solution. Do not solve for the undetermined coefficients in your proposed solution.

$$(i) \quad y'' - 2y' + y = 3x(x - 4),$$

$$(ii) \quad y'' + 4y = 16x \cos 2x,$$

$$(iii) \quad y'' - y' - 2y = 40 \sin^2 x,$$

$$(iv) \quad y'' - 4y = 2 \cos x + 2e^{2x}.$$

**Solution:**

$$(i) \quad y_p(x) = A_0 + A_1 x + A_2 x^2$$

$$(ii) \quad y_p(x) = x((A_0 + A_1 x) \cos 2x + (B_0 + B_1 x) \sin 2x)$$

$$(iii) \quad y_p(x) = A + (B_0 \cos 2x + B_1 \sin 2x),$$

$$(iv) \quad y_p(x) = A_0 \cos x + B_0 \sin x + A x e^{2x}$$

**C) Method of Variation-of-Parameters**

Use the variation-of-parameters method to find the general solution to the given DE.

$$(i) \quad y'' + 6y' + 9y = \frac{2e^{-3x}}{x^2+1},$$

$$(ii) \quad y'' - 2y' + y = 4e^x x^{-3} \ln x, \quad x > 0,$$

$$(iii) \quad y'' + 4y' + 4y = 15e^{-2x} \ln x + 25 \cos x, \quad x > 0,$$

**Solution:**

$$(i) \quad y(x) = e^{-3x} (c_1 + c_2 x + 2x \tan^{-1} x - \ln(x^2 + 1))$$

$$(ii) \quad y(x) = e^x (c_1 + c_2 x + x^{-1}(2 \ln x + 3))$$

$$(iii) \quad y(x) = c_1 e^{-2x} + c_2 x e^{-2x} + 3 \cos x + 4 \sin x + \frac{15}{4} x^2 e^{-2x} (2 \ln x - 3)$$